

005 Week 2 — English Worksheet (Intermediate)

Topic: One Function, Many Booths — Arguments & Parameters · **Course:** Theme Park Creator · **Time:** about 45 minutes

This week you build a park **booth** with a **function**. You give it **parameters** — the materials and the size it should use — so the **same** function can build a wood booth, a stone booth, a big one or a small one, just by changing the **arguments** you **call** it with.

These are the blocks you use this week:

```
blocks.fill(OAK_PLANKS, pos(0, 0, 0), pos(4, 0, 4), FillOperation.HOLLOW)
blocks.fill(STONE, pos(0, 0, 0), pos(4, 0, 4), FillOperation.REPLACE)
blocks.fill(AIR, pos(2, 0, 0), pos(2, 1, 0), FillOperation.REPLACE)
```

`blocks.fill(material, corner1, corner2, ...)` fills a box between two corners. `pos(x, y, z)` is a spot: `x` = left/right, `y` = up, `z` = forward/back. `FillOperation.HOLLOW` = only the outer **walls**. `FillOperation.REPLACE` = the **whole solid** box. `AIR` = empty space. Fill a slot with `AIR` to **carve a doorway** out of a wall.

1 · Meet Arguments and Parameters 🎁

A **parameter** is an empty box with a name — you make it when you *define* the function. An **argument** is the real value you drop into that box — you send it when you *call* the function. They line up **by position**, left to right.

MAKE IT → the names in () are PARAMETERS (empty named boxes):

```
def build_booth( wall , floor , size ):
```

CALL IT → the values in () are ARGUMENTS (real values sent in):

```
build_booth( OAK_PLANKS , STONE , 5 )
```

Position matches – 1st→1st, 2nd→2nd, 3rd→3rd:

```
wall ← OAK_PLANKS      (1st box gets the 1st value)
```

```
floor ← STONE           (2nd box gets the 2nd value)
```

```
size ← 5                (3rd box gets the 3rd value)
```

Why does one function with parameters let you build many different booths?

2 · Predict 🧠

Read each function. Before you imagine the booth, write what you think happens.

```
def build_booth(wall, floor, size):  
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)  
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)  
  
build_booth(OAK_PLANKS, STONE, 5)
```

What material is the floor? What material are the walls? How wide is the booth?


```
def build_booth(wall, floor, size):  
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)  
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)  
  
build_booth(GLASS, STONE, 3)  
build_booth(BRICKS, OAK_PLANKS, 8)
```

The same function is called twice with different arguments. Are the two booths the same? Which one is bigger?

3 · Spot the Bug 🐛

Each block was meant to do something but is broken. Read what it is **supposed** to do, rewrite it so it works, then explain why the original was wrong.

Bug A — This should build one booth. But when you run it, **nothing happens at all**.

```
def build_booth(wall, floor, size):  
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)  
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)
```

Hint: writing `def` only **teaches** the function. Something is missing after it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The booth should be as wide as the number you send in. But `build_booth(OAK_PLANKS, STONE, 9)` and `build_booth(OAK_PLANKS, STONE, 2)` build the **same size** every time.

```
def build_booth(wall, floor, size):  
    blocks.fill(floor, pos(0, 0, 0), pos(4, 0, 4), FillOperation.REPLACE)  
    blocks.fill(wall, pos(0, 0, 0), pos(4, 3, 4), FillOperation.HOLLOW)  
  
build_booth(OAK_PLANKS, STONE, 9)
```

Hint: look at the numbers inside. The function never uses `size`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should build a **stone floor** with **glass walls**. But it comes out with glass on the floor and stone walls — the wrong way around.

```
def build_booth(wall, floor, size):  
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)  
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)  
  
build_booth(STONE, GLASS, 5)
```

Hint: the arguments line up with the parameters **by position**. Which value landed in which box?

Write the fixed code:

Why was it wrong? Why does your fix work?

Now the real challenge. In your homework world, write **one** function called `build_booth` that builds a booth a visitor can walk into. Your function must have these **three parameters**:

```
def build_booth(wall, floor, size):
```

Your booth must include:

- a **floor** made from the `floor` material, `size` wide
- **walls** made from the `wall` material (use `FillOperation.HOLLOW` so the middle stays empty)
- a **doorway** carved out of one wall with `AIR` so a visitor can walk in

Then **call your function twice** with different materials and sizes, for example:

```
build_booth(OAK_PLANKS, STONE, 5)
build_booth(BRICKS, OAK_PLANKS, 8)
```

Write your finished `build_booth` function and the two lines that call it:

5 · Tell Me What You Built 📷

When your two booths stand side by side, send a photo or video and explain what you did. Use these sentence starters — write 4 to 6 sentences.

My three parameters were ...

The argument that changed the booth the most was ...

To carve the doorway I ...

The same function built two different booths because ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) of your code and your booths. Talk it through like you are teaching someone who has never seen it. Try to use these words: **function**, **parameter**, **argument**, **call**, **position**.

1. Show your `build_booth` function and read the line with the three **parameters** out loud.
2. Point at one parameter and say what it controls.
3. Run a **call** and show the booth appear, then run another with different **arguments**.
4. Say in your own words how parameters and arguments line up by **position**.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.